

CI/CD Pipeline for Brain Tasks Application using AWS Services

Aim

To build and deploy a containerized Brain Tasks application using AWS CodeBuild, Amazon ECR, Amazon EKS, Kubernetes, Docker, and CloudWatch.

Tools Used

- GitHub
 - Docker
 - AWS CodeBuild
 - Amazon ECR
 - Amazon EKS
 - Kubernetes
 - CloudWatch
 - kubectl
-

Architecture Diagram

GitHub Repository



AWS CodePipeline



AWS CodeBuild



Docker Image Build



Amazon ECR



Amazon EKS



Kubernetes Deployment



LoadBalancer Service



Brain Tasks Application

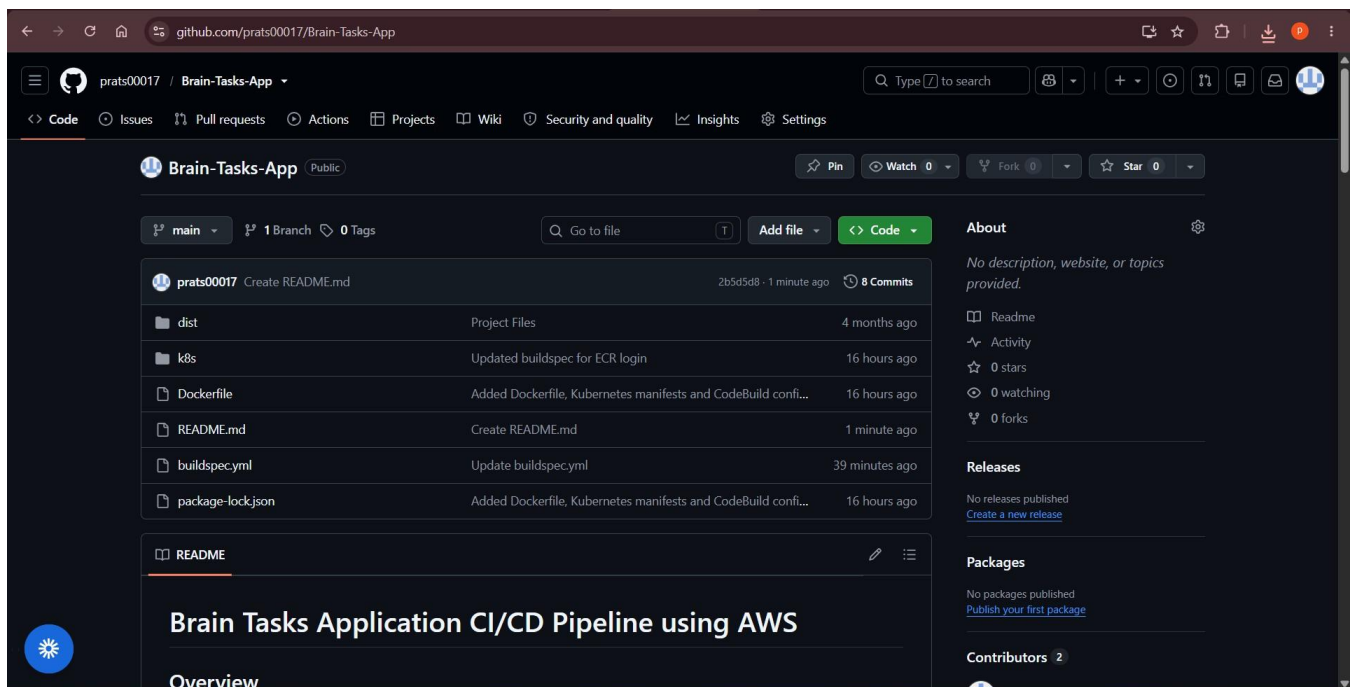


CloudWatch Monitoring

Step 1: Source Code Repository

The Brain Tasks application source code was maintained in GitHub.

Screenshot:



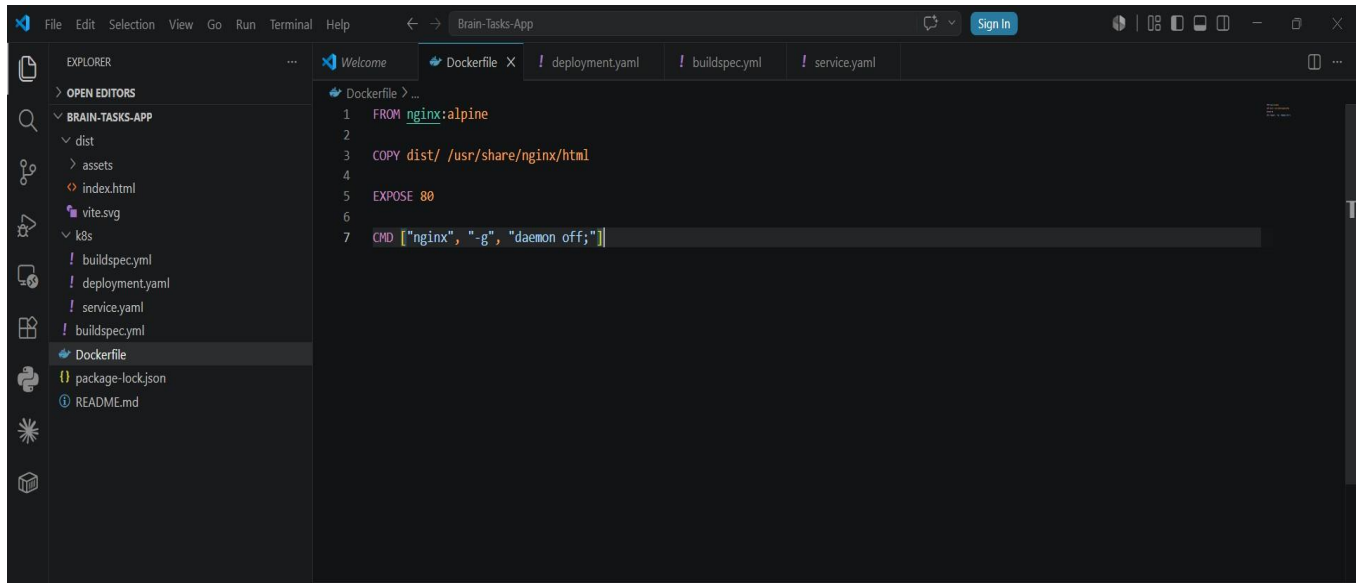
Step 2: Dockerization

A Dockerfile was created to containerize the application.

Command:

```
docker build -t brain-tasks-app .
```

Screenshot:



Step 3: AWS CodeBuild

Created CodeBuild project:

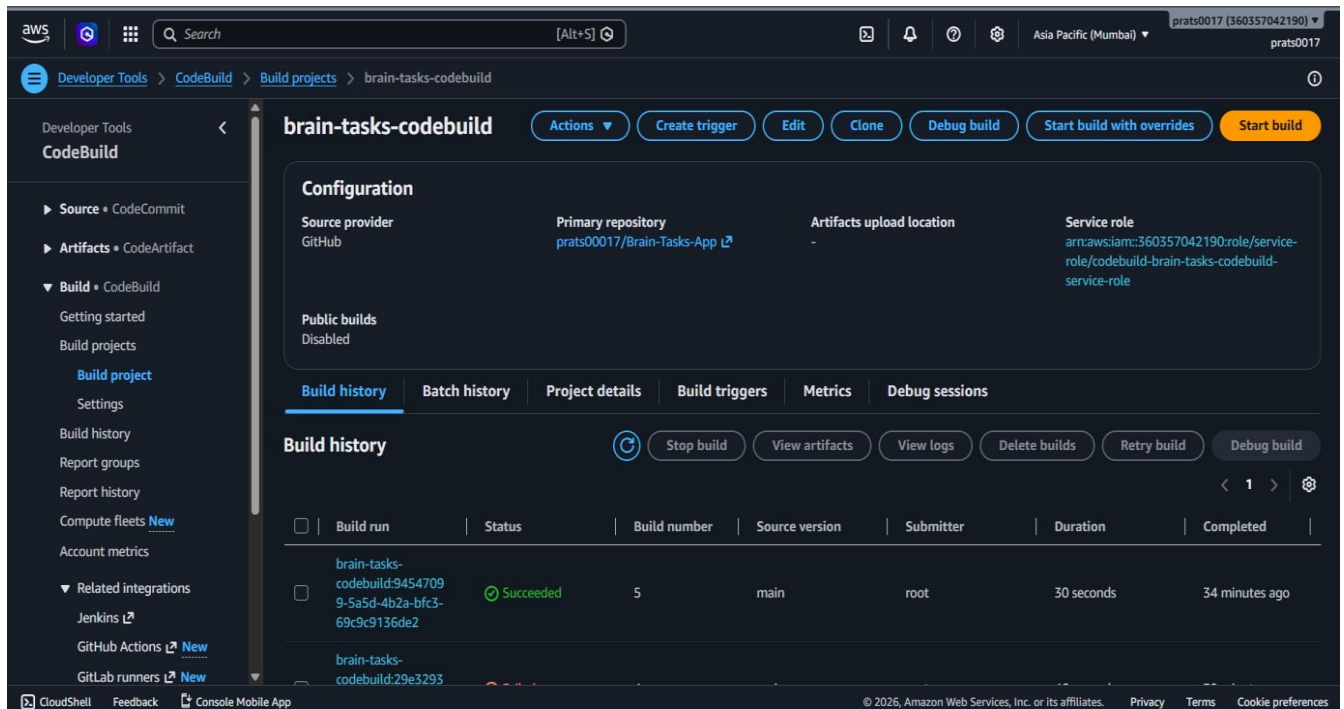
brain-tasks-codebuild

Configured:

- Source repository integration with GitHub.
- buildspec.yml file for automated build instructions.
- Docker image build process.
- Integration with AWS CodePipeline.

Build execution completed successfully and artifacts were generated.

Screenshot:



Step 4: AWS CodePipeline

Created pipeline:

brain-tasks-pipeline

Configured stages:

1. Source Stage
 - GitHub Repository
2. Build Stage
 - AWS CodeBuild

Pipeline execution was completed successfully.

This automated the Continuous Integration process and ensured that source code changes trigger the build workflow.

Screenshots:

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Developer Tools

CodePipeline

Pipelines

brain-tasks-pipeline

brain-tasks-pipeline

Edit

Stop execution

Create trigger

Clone pipeline

Release change

Pipeline

Executions

Triggers

Settings

Tags

Source

260bd312-768f-4366-84a1-d8d70fb3b6f8

All actions succeeded.

Source

GitHub (via GitHub App)

1 minute ago

0f89d6d4 Source: Add files via upload

Build

260bd312-768f-4366-84a1-d8d70fb3b6f8

All actions succeeded.

Build

AWS CodeBuild

1 minute ago

0f89d6d4 Source: Add files via upload

CloudShell

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Developer Tools

CodePipeline

Pipelines

brain-tasks-pipeline

Success

Congratulations! The pipeline brain-tasks-pipeline has been created.

brain-tasks-pipeline

Edit

Stop execution

Create trigger

Clone pipeline

Release change

Pipeline

Executions

Triggers

Settings

Tags

Execution ID

Status

Source revisions

Trigger

Started

Duration

Completed

260bd312

Succeeded

Source - 0f89d6d4 Add files via upload

CreatePipeline - root

Jun 24, 2026 10:51 AM (UTC+5:30)

1 minute 14 seconds

Jun 24, 2026 10:52 AM (UTC+5:30)

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Developer Tools > CodePipeline > Pipelines

Developer Tools
CodePipeline

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- Artifacts • CodeArtifact
- Build • CodeBuild
- Deploy • CodeDeploy
- Pipeline • CodePipeline
 - Getting started
 - Pipelines
 - Account metrics
- Settings

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Feedback

Pipelines info

View history Release change Delete pipeline Create pipeline

Q

	Name	Latest execution status	Latest source revisions	Latest execution started	Most recent executions
	brain-tasks-pipeline	Succeeded	Source - 0f89d6d4: Add files via upload	7 minutes ago	View details

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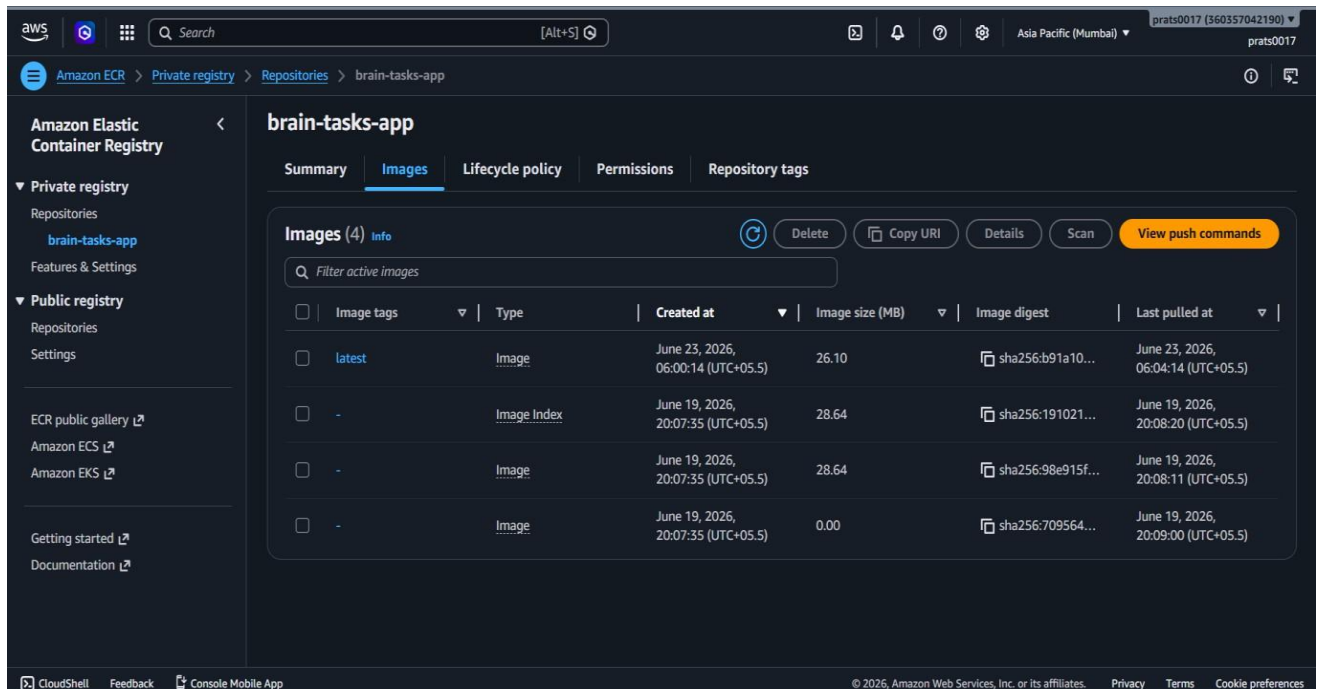
Step 5: Amazon ECR

Repository:

brain-tasks-app

Docker image was pushed successfully.

Screenshot:



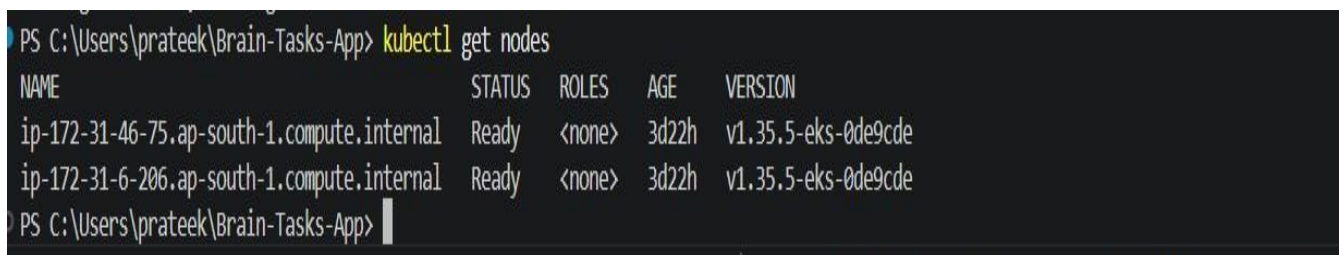
Step 6: Amazon EKS

Cluster nodes verified.

Command:

```
kubectl get nodes
```

Screenshot:



Step 7: Deployment

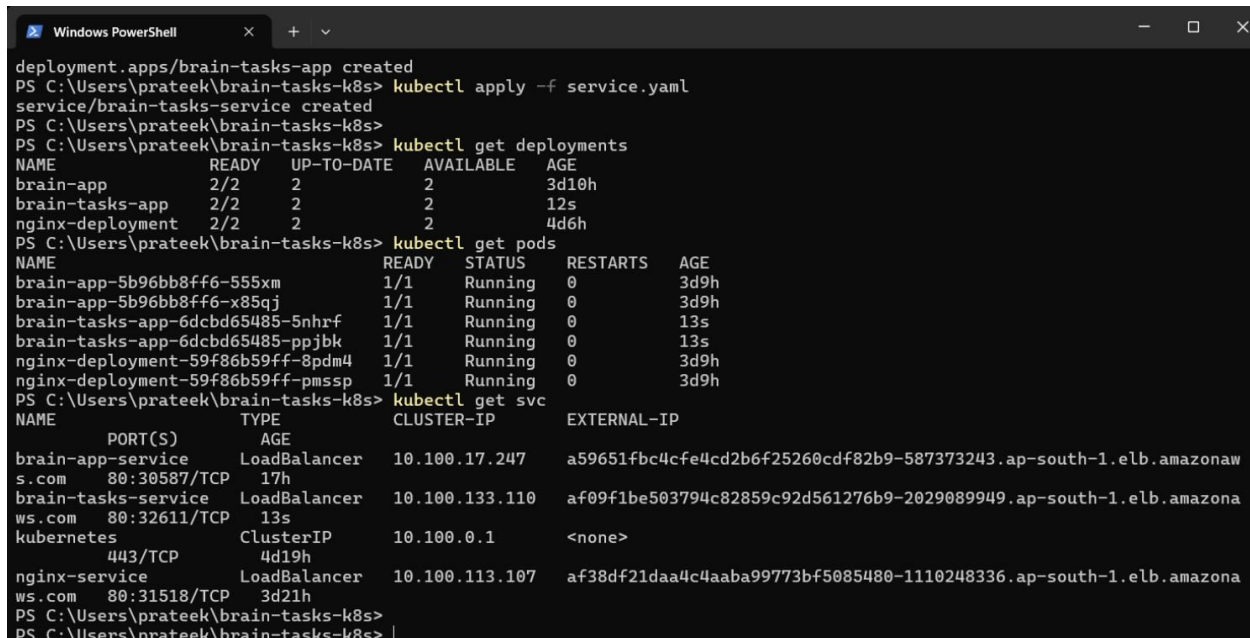
Applied Kubernetes manifests.

Commands:

```
kubectl apply -f deployment.yaml
```

```
kubectl apply -f service.yaml
```

Screenshot:



```
deployment.apps/brain-tasks-app created
PS C:\Users\prateek\brain-tasks-k8s> kubectl apply -f service.yaml
service/brain-tasks-service created
PS C:\Users\prateek\brain-tasks-k8s>
PS C:\Users\prateek\brain-tasks-k8s> kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
brain-app            2/2     2             2           3d10h
brain-tasks-app      2/2     2             2           12s
nginx-deployment     2/2     2             2           4d6h
PS C:\Users\prateek\brain-tasks-k8s> kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
brain-app-5b96bb8ff6-555xm    1/1     Running   0           3d9h
brain-app-5b96bb8ff6-x85qj    1/1     Running   0           3d9h
brain-tasks-app-6dcdb65485-5nhrf  1/1     Running   0           13s
brain-tasks-app-6dcdb65485-ppjbj  1/1     Running   0           13s
nginx-deployment-59f86b59ff-8pdm4  1/1     Running   0           3d9h
nginx-deployment-59f86b59ff-pmssp  1/1     Running   0           3d9h
PS C:\Users\prateek\brain-tasks-k8s> kubectl get svc
NAME                TYPE          CLUSTER-IP      EXTERNAL-IP
brain-app-service   LoadBalancer 10.100.17.247    a59651fbc4cfe4cd2b6f25260cdf82b9-587373243.ap-south-1.elb.amazonaws.com
brain-tasks-service LoadBalancer 10.100.133.110   af09f1be503794c82859c92d561276b9-2029089949.ap-south-1.elb.amazonaws.com
kubernetes          ClusterIP      10.100.0.1       <none>
nginx-service       LoadBalancer 10.100.113.107   af38df21daa4c4aaba99773bf5085480-1110248336.ap-south-1.elb.amazonaws.com
PS C:\Users\prateek\brain-tasks-k8s>
PS C:\Users\prateek\brain-tasks-k8s> |
```

Step 8: Pods Verification

Command:

```
kubectl get pods
```

Screenshot:

As above

Step 9: Services Verification

Command:

```
kubectl get svc
```

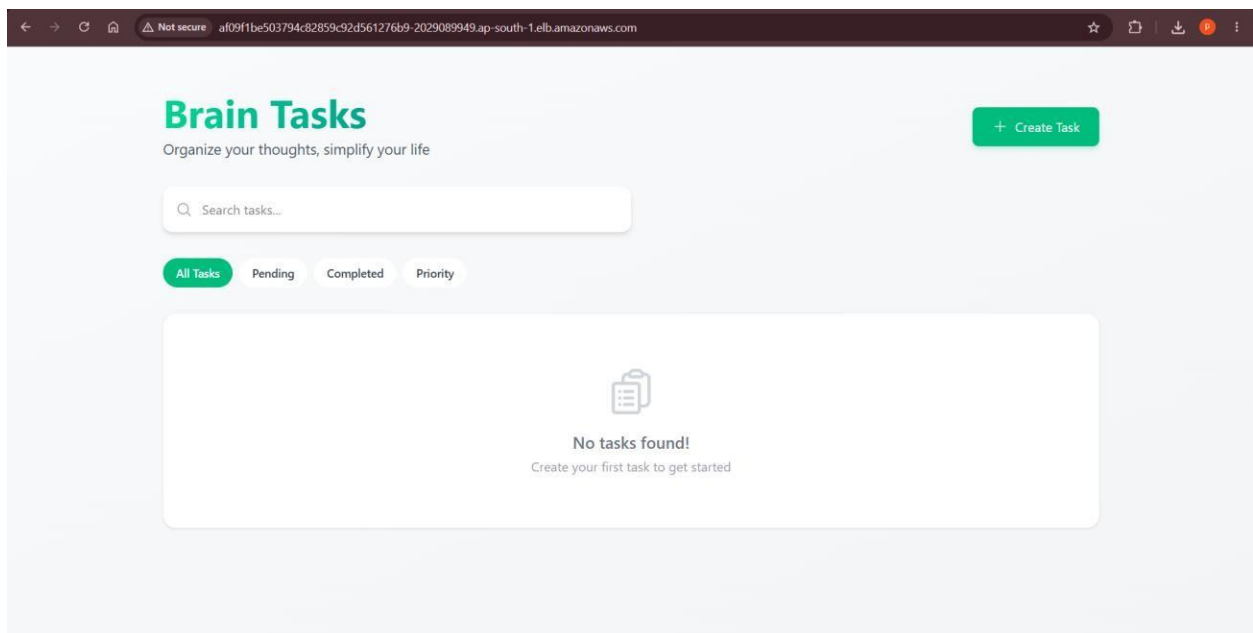
Screenshot:

As above

Step 10: Application Output

Application was accessed through the external LoadBalancer URL.

Screenshot:



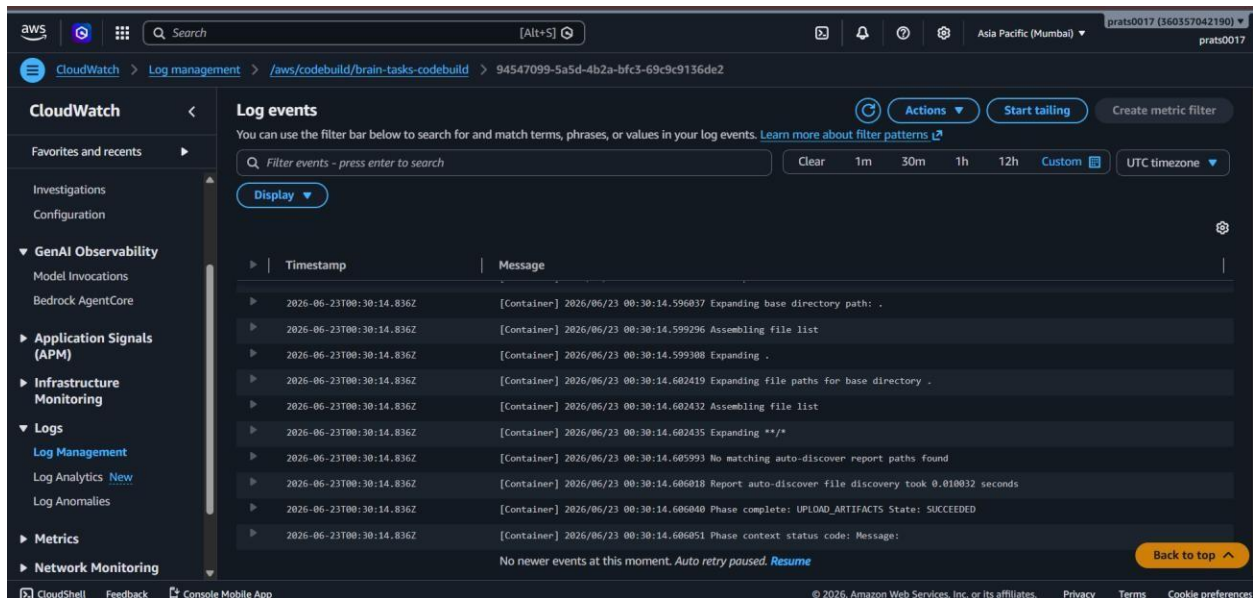
Step 11: CloudWatch Monitoring

CloudWatch log group:

```
/aws/codebuild/brain-tasks-codebuild
```

Build logs were monitored successfully.

Screenshot:



Commands Used

```
kubectl get nodes
kubectl get deployments
kubectl get pods
kubectl get svc
kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
```

Result

Successfully implemented and deployed the Brain Tasks application using a complete CI/CD pipeline with GitHub, AWS CodePipeline, AWS CodeBuild, Amazon ECR, Amazon EKS, Kubernetes and CloudWatch.

The application was containerized using Docker, images were stored in Amazon ECR, deployed on Amazon EKS, and exposed externally through a LoadBalancer service. Monitoring and build activities were tracked using CloudWatch.

Conclusion

The project demonstrates an end-to-end DevOps workflow using AWS cloud-native services. Continuous Integration was achieved through AWS CodePipeline and AWS CodeBuild, while Continuous Deployment was performed using Kubernetes on Amazon EKS.

The Brain Tasks application was successfully deployed and made accessible through a LoadBalancer service. Monitoring and logging were achieved using Amazon CloudWatch, providing a complete CI/CD implementation.

